

MediNotes: Agentic Healthcare SaaS

Multi-agent clinical documentation platform that converts multimodal visit data into structured, verifiable summaries.

PROBLEM

Clinicians spend significant time turning scattered inputs (handwritten notes, audio, and uploaded files) into consistent documentation. Manual workflows increase risk of omissions, low traceability, and reduced time for patient care.

SOLUTION FEATURES

- Multimodal intake: PDF/DOCX/TXT, audio transcription (Whisper), image OCR.
- Live summary generation with SSE streaming and resumable jobs.
- Critic QA loop with regeneration to reduce hallucinations and missing facts.
- Evidence mapping ties claims back to source snippets for auditability.
- Patient memory (RAG) for longitudinal continuity and context recall.
- MediNotes Assistant chat co-pilot for context-aware clinical Q&A.
- Agentic email dispatch with optional translation and delivery via Resend.

ARCHITECTURE

1. Next.js frontend sends consultation data to FastAPI (`/api/consultation`).
2. Summary Agent orchestrates: Extraction -> Research (MCP) -> Critic -> Evidence -> Memory.
3. Coordinator Agent extracts structured next actions from generated notes.
4. MediNotes Assistant uses `/api/chat` for context-aware Q&A over current summary + patient history.
5. Events stream back to UI in real time (`status`, `actions`, `evidence`, `summary`).
6. Persistence stores summaries/notes/evidence for future retrieval and chat context.

Hub-and-spoke agent topology

MCP external research boundary

Fallback model routing

TECHNOLOGY STACK

Frontend	Next.js 16, React 19, Clerk auth, Server-Sent Events (SSE)
Backend	FastAPI, Python 3.12, Uvicorn, Pydantic
AI/Agents	OpenAI SDK, OpenAI Agents SDK (MCP), DeepSeek/Gemini/OpenAI routing, Whisper
Data/State	JSON vector memory store (`text-embedding-3-small`), optional Upstash Redis job store
Integrations	Brave Search MCP server (guidelines/interactions), Resend email API
Deployment	Docker multi-stage build, AWS ECR, AWS App Runner